

Harbour Tech Ltd: Small Office Network

Cisco Packet Tracer project log | Leonardo Vecchia | September–October 2026

Introduction

Purpose

This project was built as hands-on practice ahead of my networking internship search, alongside my third year of Networking Technologies at TU Dublin and the CCNA preparation courses I am taking. The goal was to design, build, secure and test a small multi-VLAN office network from scratch, and to document every step, including the problems I ran into and how I solved them.

The scenario

To give the build a realistic brief, I set up a fictional company, Harbour Tech Ltd: a small office with staff workstations, a guest area for visitors and an internal file server. The network had to meet these requirements:

- Separate Staff, Guest and Server traffic into their own VLANs
- Route between VLANs through a single router (router-on-a-stick)
- Hand out addresses to staff and guest devices automatically with DHCP, keeping a reserved range for static devices
- Stop guests from reaching the server network while still allowing everything else
- Manage all network devices securely over SSH from a dedicated management VLAN
- Harden the switches: shut down unused ports and keep no traffic untagged on trunks

Skills demonstrated

VLANs and 802.1Q trunking, VTP, inter-VLAN routing, DHCP, extended ACLs, SSH, basic switch hardening, and structured testing and troubleshooting.

Step 1: Topology and Addressing Plan

Devices

- R1: Cisco 2911 router (router-on-a-stick)
- SW1 and SW2: Cisco 2960 switches
- SW1: 2 Staff PCs (Staff 0, Staff 1) and the Server
- SW2: 1 Staff PC (Staff 2) and 2 Guest laptops (Guest 1, Guest 2)

Staff PCs were placed on both switches on purpose, to prove that VLAN 10 is carried correctly across the trunk.

Physical links

Link	Interfaces	Type
R1 – SW1	R1 G0/0 → SW1 G0/1	802.1Q trunk
SW1 – SW2	SW1 G0/2 → SW2 G0/1	802.1Q trunk
SW1 – Staff 0, Staff 1, Server	Fa0/1, Fa0/2, Fa0/3	Access
SW2 – Staff 2, Guest 1, Guest 2	Fa0/1, Fa0/2, Fa0/3	Access

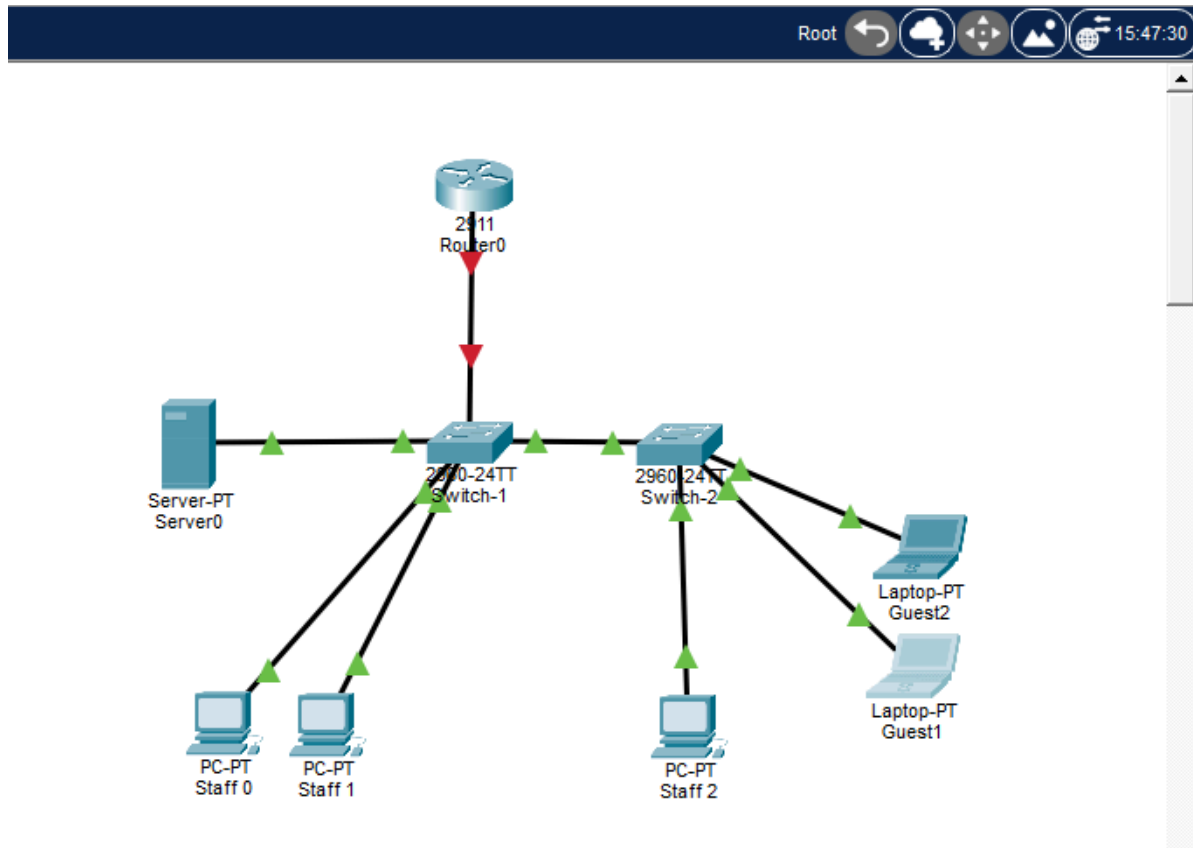


Figure 1: Harbour Tech topology in Packet Tracer (R1 link shown red before the router interface was enabled)

Addressing plan

VLAN	Name	Subnet	Gateway (R1)	Addressing
10	STAFF	10.10.10.0/24	10.10.10.1 (G0/0.10)	DHCP (.21–.254)
20	GUEST	10.10.20.0/24	10.10.20.1 (G0/0.20)	DHCP (.21–.254)
30	SERVER	10.10.30.0/24	10.10.30.1 (G0/0.30)	Static: Server 10.10.30.10
99	MGMT	10.10.99.0/24	10.10.99.1 (G0/0.99)	Static: SW1 .11, SW2 .12
999	NATIVE_UNUSED	none	none	Native VLAN on trunks only

Addresses .1–.20 in each DHCP subnet are reserved for the gateway and future static devices such as printers and access points.

Step 2: Switch Configuration (Layer 2)

2.1 Baseline settings

```
SW1(config)# no ip domain-lookup
SW2(config)# no ip domain-lookup
R1(config)# no ip domain-lookup
```

DNS lookup was disabled on all devices so a mistyped command is not treated as a hostname, which freezes the CLI for several seconds.

2.2 VTP

SW1 was set as VTP server and SW2 as client, so VLANs created on SW1 propagate automatically to SW2. This keeps the VLAN databases consistent and makes future expansion easier.

```
SW1(config)# vtp domain SWTSW
SW1(config)# vtp mode server
SW1(config)# vtp password <password>

SW2(config)# vtp domain SWTSW
SW2(config)# vtp mode client
SW2(config)# vtp password <password>
```

Risk considered: A new switch joining the domain with a higher configuration revision number can overwrite the VLAN database on every switch. A VTP password was set to reduce this risk. VTPv3 would be a stronger option, as it introduces a primary server role that controls who can make changes.

2.3 VLAN creation

VLANs were created on SW1 only, since it is the VTP server.

```
SW1(config)# vlan 10
SW1(config-vlan)# name STAFF
SW1(config-vlan)# vlan 20
SW1(config-vlan)# name GUEST
SW1(config-vlan)# vlan 30
SW1(config-vlan)# name SERVER
SW1(config-vlan)# vlan 99
SW1(config-vlan)# name MGMT
SW1(config-vlan)# vlan 999
SW1(config-vlan)# name NATIVE_UNUSED
```

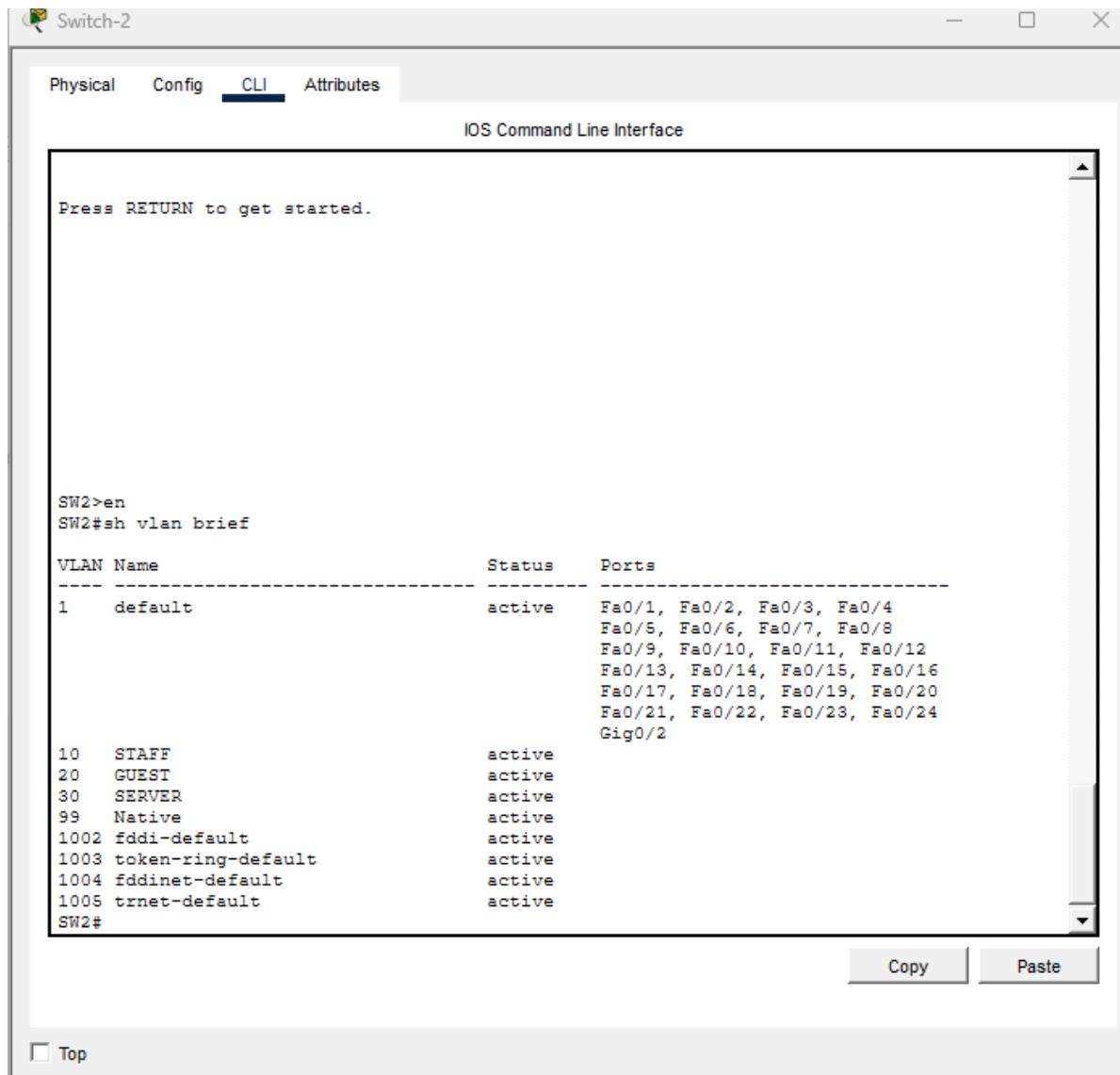


Figure 2: SW2 VLAN database received via VTP (captured before access ports were assigned and before VLAN 99 was renamed)

Design change: VLAN 99 was first named "Native". It was later renamed MGMT and dedicated to device management, and an unused VLAN 999 was created as the native VLAN. This way management traffic is always tagged and nothing important crosses the trunks untagged.

2.4 Trunks and native VLAN

```
SW1(config)# interface range g0/1 - 2
SW1(config-if-range)# switchport mode trunk
SW1(config-if-range)# switchport trunk native vlan 999

SW2(config)# interface g0/1
SW2(config-if)# switchport mode trunk
SW2(config-if)# switchport trunk native vlan 999
```

The native VLAN was set to 999 on both ends of each trunk to avoid a native VLAN mismatch. No encapsulation command was needed, since the 2960 only supports 802.1Q.

2.5 Access ports

```
SW1(config)# interface range fa0/1 - 2
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport access vlan 10
SW1(config)# interface fa0/3
SW1(config-if)# switchport mode access
SW1(config-if)# switchport access vlan 30
```

```
SW2(config)# interface fa0/1
SW2(config-if)# switchport mode access
SW2(config-if)# switchport access vlan 10
SW2(config)# interface range fa0/2 - 3
SW2(config-if-range)# switchport mode access
SW2(config-if-range)# switchport access vlan 20
```

2.6 Management VLAN

```
SW1(config)# interface vlan 99
SW1(config-if)# ip address 10.10.99.11 255.255.255.0
SW1(config-if)# no shutdown
SW1(config)# ip default-gateway 10.10.99.1
```

```
SW2(config)# interface vlan 99
SW2(config-if)# ip address 10.10.99.12 255.255.255.0
SW2(config-if)# no shutdown
SW2(config)# ip default-gateway 10.10.99.1
```

Each switch gets one IP, on VLAN 99, used only for management. The default gateway lets the switches reply to devices in other VLANs, for example an admin connecting over SSH from the Staff VLAN.

2.7 Securing unused ports

```
SW1(config)# interface range fa0/4 - 24
SW1(config-if-range)# shutdown
```

```
SW2(config)# interface range fa0/4 - 24, g0/2
SW2(config-if-range)# shutdown
```

2.8 Verification

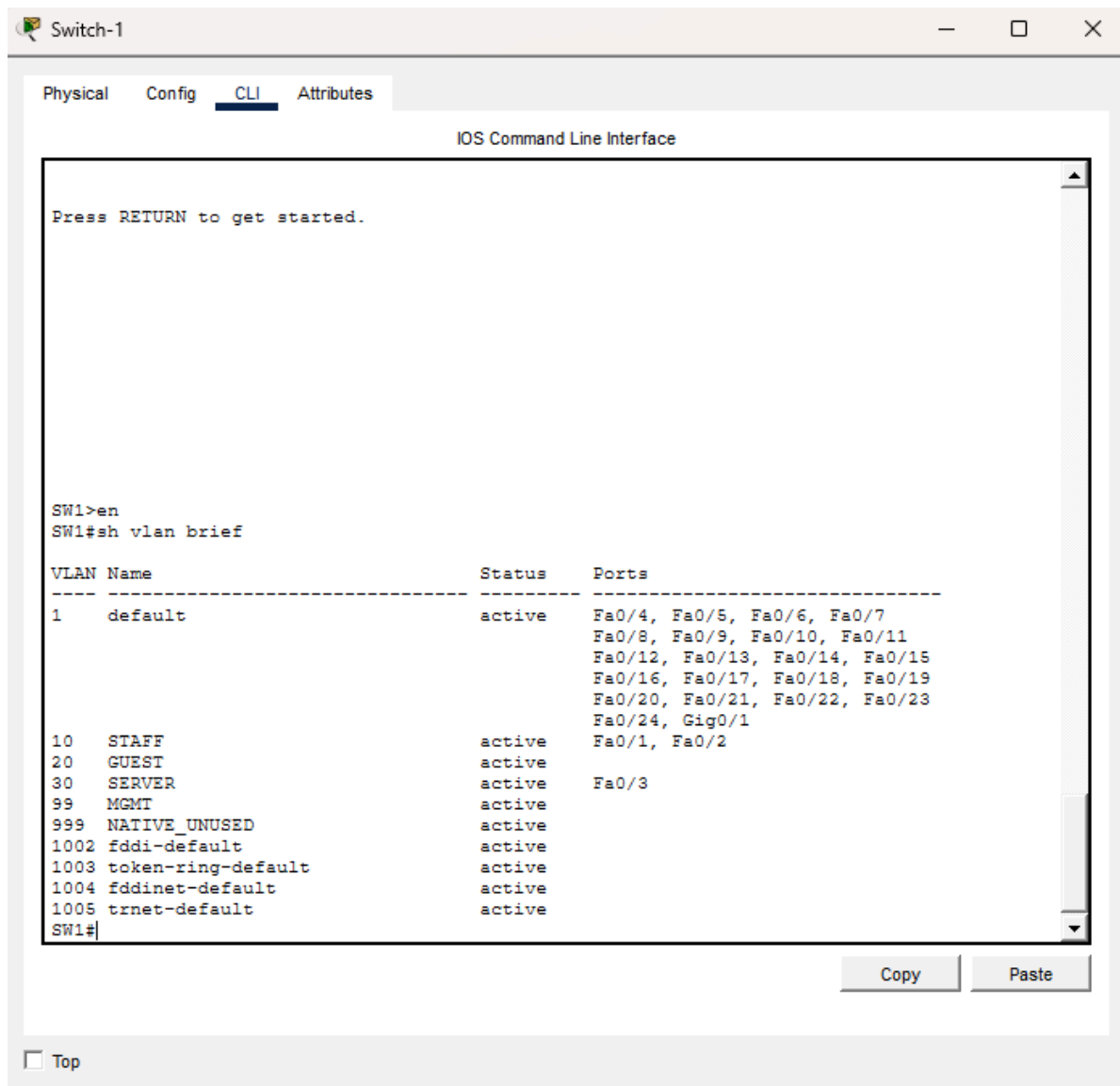


Figure 3: SW1 show vlan brief: Staff on Fa0/1–2, Server on Fa0/3, VLANs 99 and 999 renamed

```
SW1#
SW1#sh int trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/2    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/2    1,10,20,30,99,999

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/2    1,10,20,30,99,999

SW1#
```

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Figure 4: SW1 show interfaces trunk: G0/2 trunking with native VLAN 999 and 802.1Q encapsulation

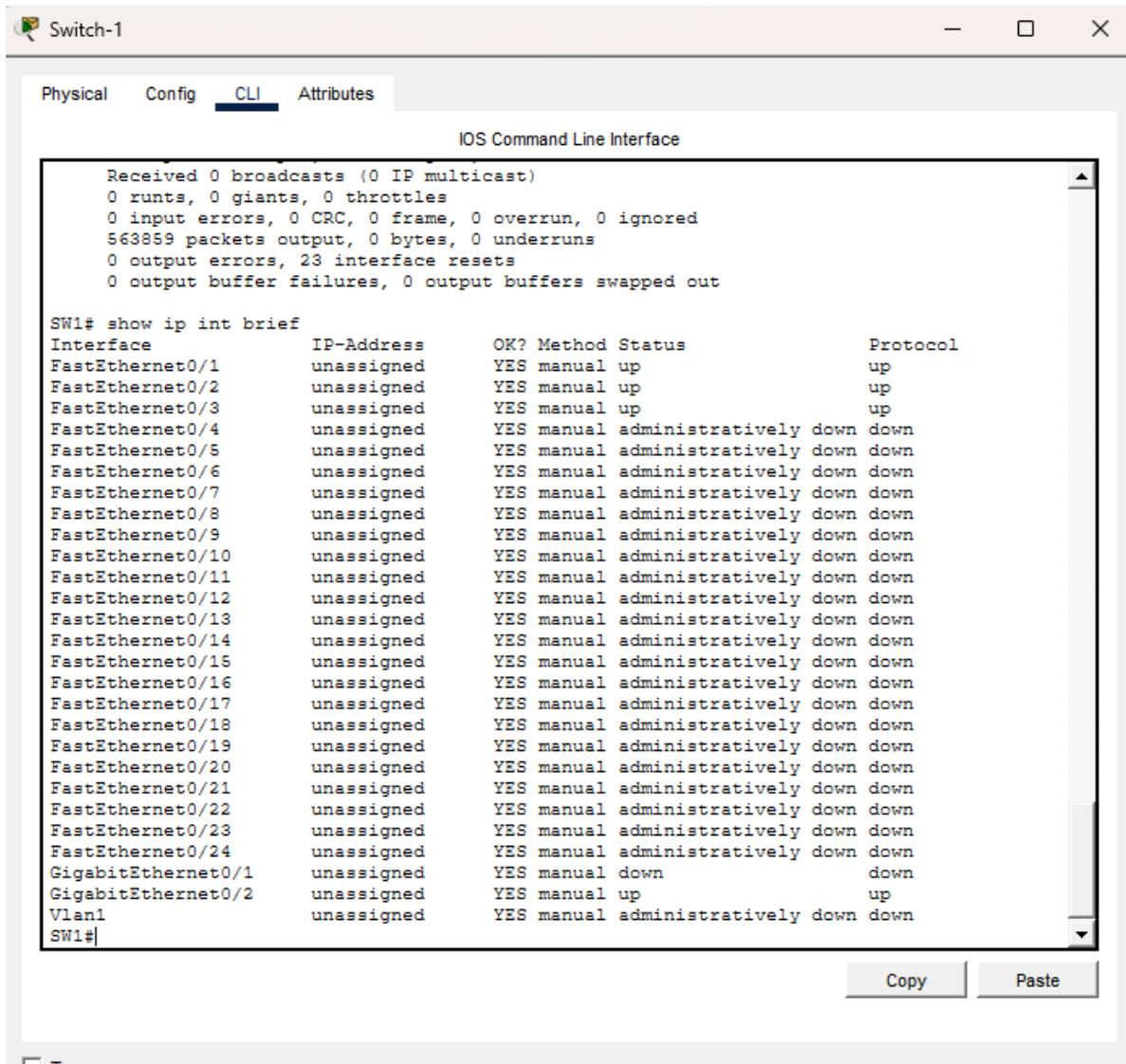


Figure 5: SW1 show ip interface brief: unused ports administratively down. G0/1 is down because R1's interface had not been enabled yet

Step 3: Inter-VLAN Routing (Router-on-a-Stick)

3.1 Configuration

```

R1(config)# interface g0/0
R1(config-if)# no shutdown

R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 10.10.10.1 255.255.255.0

R1(config)# interface g0/0.20

```

```

R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 10.10.20.1 255.255.255.0

R1(config)# interface g0/0.30
R1(config-subif)# encapsulation dot1Q 30
R1(config-subif)# ip address 10.10.30.1 255.255.255.0

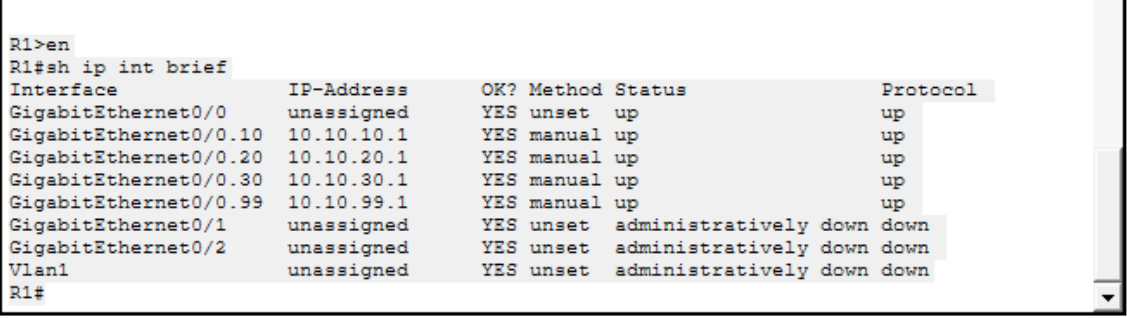
R1(config)# interface g0/0.99
R1(config-subif)# encapsulation dot1Q 99
R1(config-subif)# ip address 10.10.99.1 255.255.255.0

```

The physical interface carries no IP address; it only needs to be enabled. Each subinterface handles one VLAN, and its dot1Q tag must match the VLAN number. VLAN 999 needs no subinterface because no traffic uses it.

3.2 Verification

Each test was chosen to prove one specific link or device, so a failure points straight to the cause.



Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	10.10.10.1	YES	manual	up	up
GigabitEthernet0/0.20	10.10.20.1	YES	manual	up	up
GigabitEthernet0/0.30	10.10.30.1	YES	manual	up	up
GigabitEthernet0/0.99	10.10.99.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Figure 6: R1 show ip interface brief: G0/0 and all four subinterfaces up/up

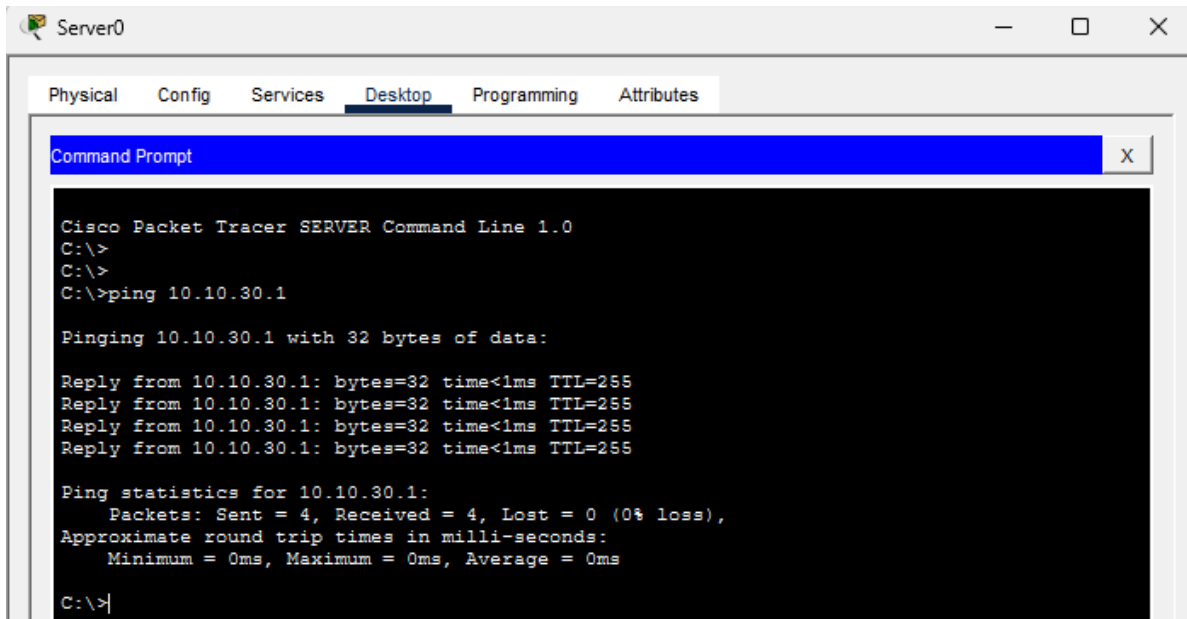


Figure 7: Server → gateway 10.10.30.1: confirms the server's VLAN 30 access port, the SW1–R1 trunk and R1's G0/0.30 subinterface

```
SW1#ping 10.10.99.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.99.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

SW1#ping 10.10.99.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.99.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

Figure 8: SW1 → gateway 10.10.99.1: confirms SW1 can reach R1 over VLAN 99, so it is manageable from other VLANs. The first lost packet is ARP resolving the router's MAC

```
SW1#ping 10.10.99.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.99.12, timeout is 2 seconds:
..!!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

SW1#ping 10.10.99.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.99.12, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

SW1#
```

Figure 9: SW1 → SW2 10.10.99.12: confirms VLAN 99 crosses the SW1–SW2 trunk (pure Layer 2, the router is not involved)

```

SW1#sh int trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    999
Gig0/2    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,99,999
Gig0/2    1,10,20,30,99,999

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,99,999
Gig0/2    1,10,20,30,99,999

SW1#

```

Figure 10: SW1 show interfaces trunk: G0/1 to R1 now trunking alongside G0/2

Step 4: DHCP

4.1 Configuration

```

R1(config)# ip dhcp excluded-address 10.10.10.1 10.10.10.20
R1(config)# ip dhcp excluded-address 10.10.20.1 10.10.20.20

R1(config)# ip dhcp pool STAFF
R1(dhcp-config)# network 10.10.10.0 255.255.255.0
R1(dhcp-config)# default-router 10.10.10.1

R1(config)# ip dhcp pool GUEST
R1(dhcp-config)# network 10.10.20.0 255.255.255.0
R1(dhcp-config)# default-router 10.10.20.1

```

Exclusions are configured before the pools so reserved addresses are never handed out. R1 picks the pool that matches the subinterface the request arrives on. VLANs 30 and 99 have no pools because servers and infrastructure use static addresses.

4.2 Issue encountered: client received a reserved address

Symptom: When Guest 2 was switched to DHCP it received 10.10.20.2, which is inside the excluded range.

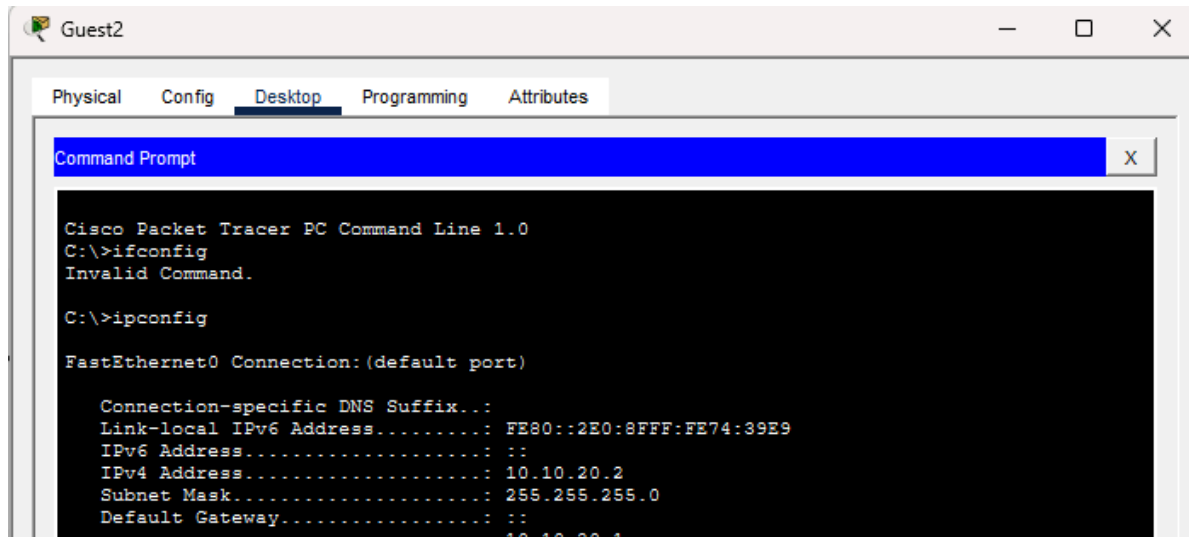


Figure 11: Guest 2 assigned 10.10.20.2 from the excluded range

Diagnosis: show running-config | include excluded showed that only the STAFF exclusion had been applied. The GUEST exclusion had not made it into the config.

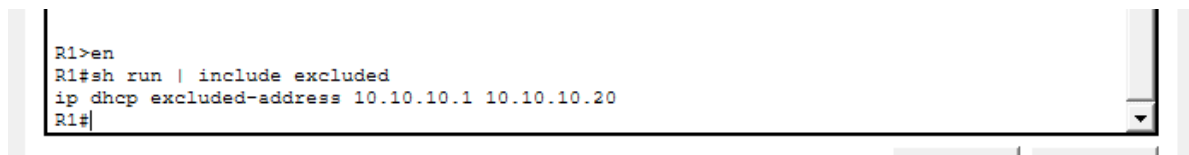


Figure 12: Only the 10.10.10.x exclusion present in the running config

Fix: Added the GUEST exclusion, cleared the bindings with clear ip dhcp binding *, and renewed the lease on the client. Guest 2 then received 10.10.20.21.

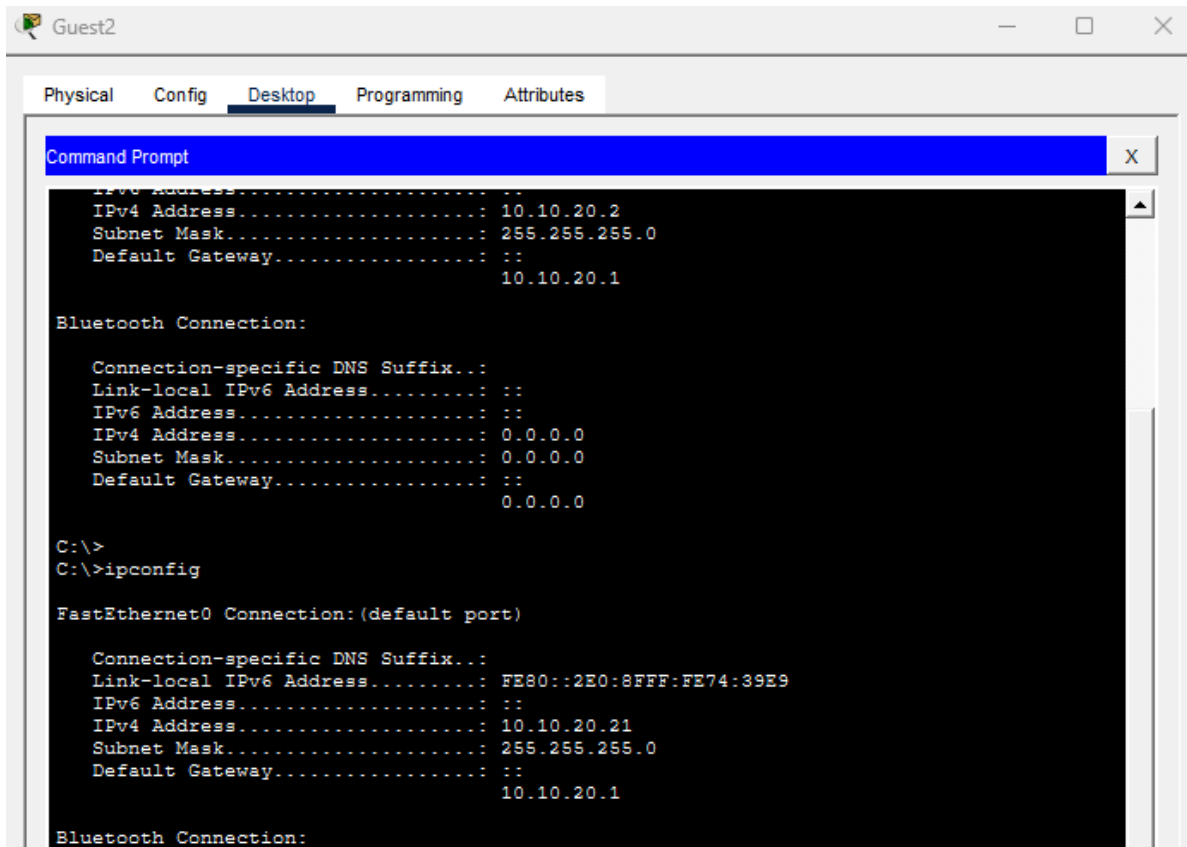


Figure 13: Guest 2 before and after renewing: 10.10.20.2 → 10.10.20.21

4.3 Verification

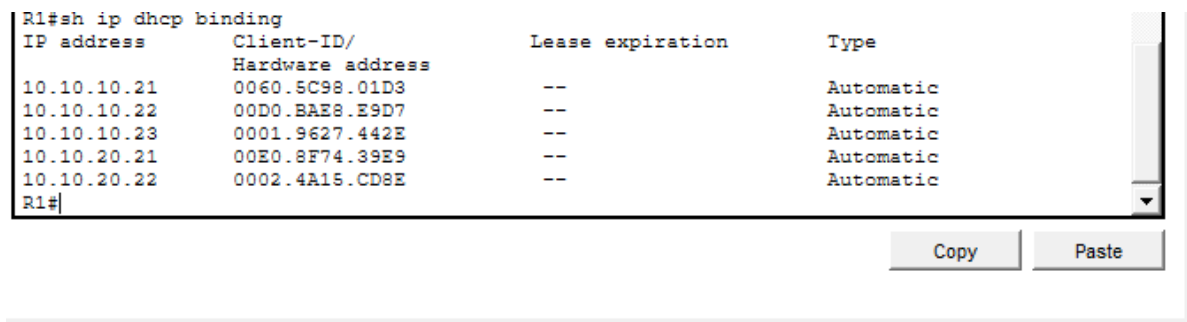


Figure 14: show ip dhcp binding: all five clients leased addresses from .21 upwards in the correct VLAN

Step 4 checkpoint: Connectivity Before Security Rules

Before adding the ACL, I confirmed every path worked, so any later change in behaviour could only come from the ACL. TTL values helped confirm the path: 128 means same VLAN (no router), 127 means one routed hop, 255 means the router itself answered.

Test	Path tested	Result	TTL
Guest 1 ↔ Guest 2	Same VLAN (20), same switch	Success	128
Staff 0 ↔ Guest 2	Inter-VLAN routing via R1	Success	127
Guest 1 → Server	Guest to Server VLAN (pre-ACL)	Success	127
Staff 2 ↔ Server	SW2 → SW1 → R1 → Server	Success	127
Staff 2 → Staff 0	Same VLAN across both switches	Success	128

```

C:\>ping 10.10.20.21

Pinging 10.10.20.21 with 32 bytes of data:

Reply from 10.10.20.21: bytes=32 time<1ms TTL=128
Reply from 10.10.20.21: bytes=32 time<1ms TTL=128
Reply from 10.10.20.21: bytes=32 time=4ms TTL=128
Reply from 10.10.20.21: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.20.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>

```

Figure 15: Guest 2 → Guest 1 (same VLAN)

```

C:\>ping 10.10.20.22

Pinging 10.10.20.22 with 32 bytes of data:

Reply from 10.10.20.22: bytes=32 time<1ms TTL=128
Reply from 10.10.20.22: bytes=32 time<1ms TTL=128
Reply from 10.10.20.22: bytes=32 time<1ms TTL=128
Reply from 10.10.20.22: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.20.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

Figure 16: Guest 1 → Guest 2 (same VLAN)

```
C:\>
C:\>ping 10.10.20.22

Pinging 10.10.20.22 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.22: bytes=32 time<1ms TTL=127
Reply from 10.10.20.22: bytes=32 time=11ms TTL=127
Reply from 10.10.20.22: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

Figure 17: Staff 0 → Guest 2. The first request timed out while ARP resolved; the remaining replies arrived

```
C:\>ping 10.10.10.23

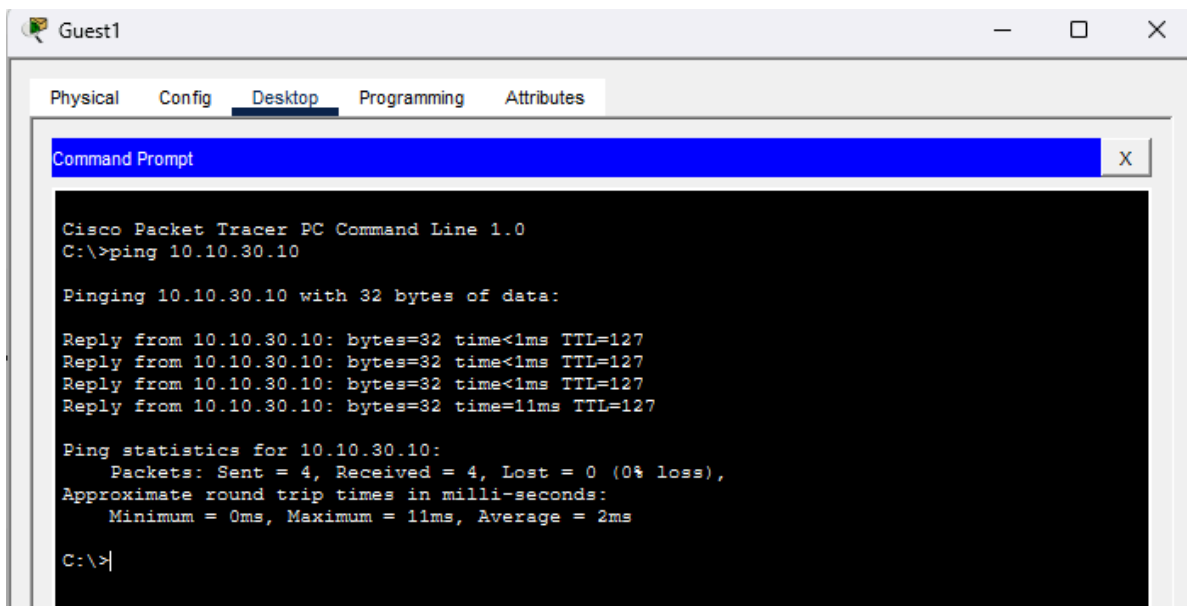
Pinging 10.10.10.23 with 32 bytes of data:

Reply from 10.10.10.23: bytes=32 time<1ms TTL=127
Reply from 10.10.10.23: bytes=32 time<1ms TTL=127
Reply from 10.10.10.23: bytes=32 time<1ms TTL=127
Reply from 10.10.10.23: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.10.23:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 18: Guest 1 → Staff 2: all four replies received



```
Guest1
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=11ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>|
```

Figure 19: Guest 1 → Server 10.10.30.10 succeeds before the ACL. TTL 127 shows the reply came from the server one hop away

```
C:\>ping 10.10.20.22

Pinging 10.10.20.22 with 32 bytes of data:

Reply from 10.10.20.22: bytes=32 time<1ms TTL=127
Reply from 10.10.20.22: bytes=32 time<1ms TTL=127
Reply from 10.10.20.22: bytes=32 time=1ms TTL=127
Reply from 10.10.20.22: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 20: Server → Guest 2

```
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=13ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

Figure 21: Staff 2 (SW2) → Server, crossing both switches and the router

```
C:\>ping 10.10.10.21

Pinging 10.10.10.21 with 32 bytes of data:

Reply from 10.10.10.21: bytes=32 time<1ms TTL=127
Reply from 10.10.10.21: bytes=32 time=11ms TTL=127
Reply from 10.10.10.21: bytes=32 time=1ms TTL=127
Reply from 10.10.10.21: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

Figure 22: Server → Staff 0

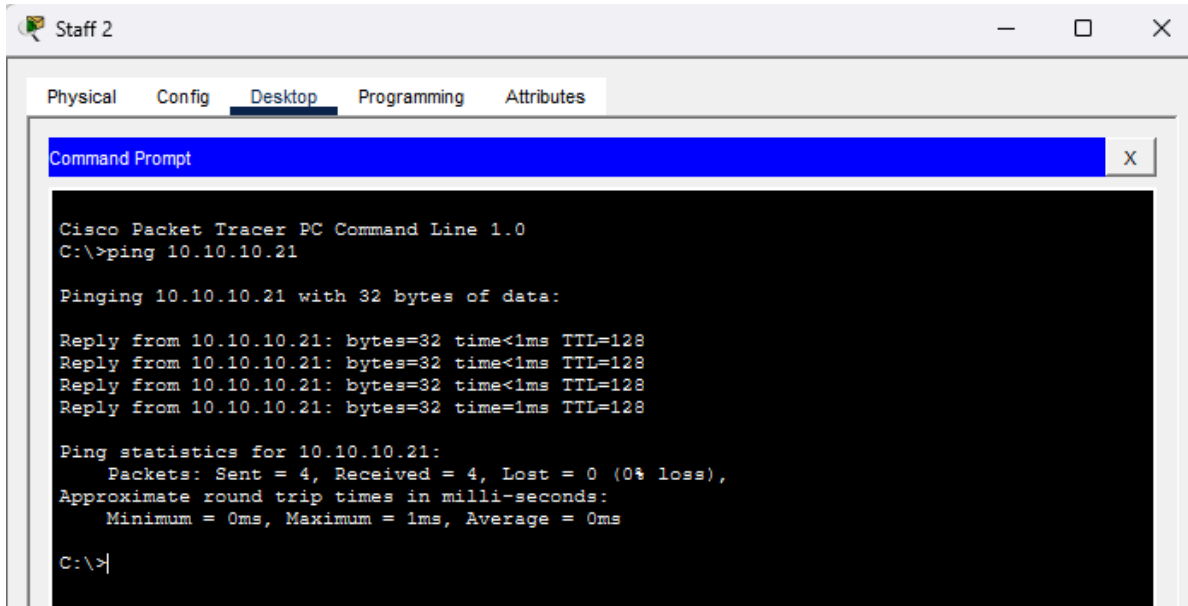


Figure 23: Staff 2 → Staff 0: VLAN 10 is carried over the SW1–SW2 trunk; traffic stays at Layer 2 (TTL 128)

Step 5: Access Control List

5.1 Goal and design decision

Guests are the least trusted users on the network, so they should not be able to reach the server network. Everything else keeps working.

Placement: The extended ACL is applied inbound on G0/0.20, closest to the source, so guest traffic to the server VLAN is dropped as soon as it enters R1, before any routing takes place. Applying it outbound on G0/0.30 instead would make R1 route the packet first, and it would inspect all traffic heading to the servers, including staff traffic.

5.2 Configuration

```
R1(config)# ip access-list extended GUEST_BLOCKER
R1(config-ext-nacl)# remark Block Guest VLAN from Server VLAN
R1(config-ext-nacl)# deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
R1(config-ext-nacl)# permit ip any any
R1(config)# interface g0/0.20
R1(config-subif)# ip access-group GUEST_BLOCKER in
```

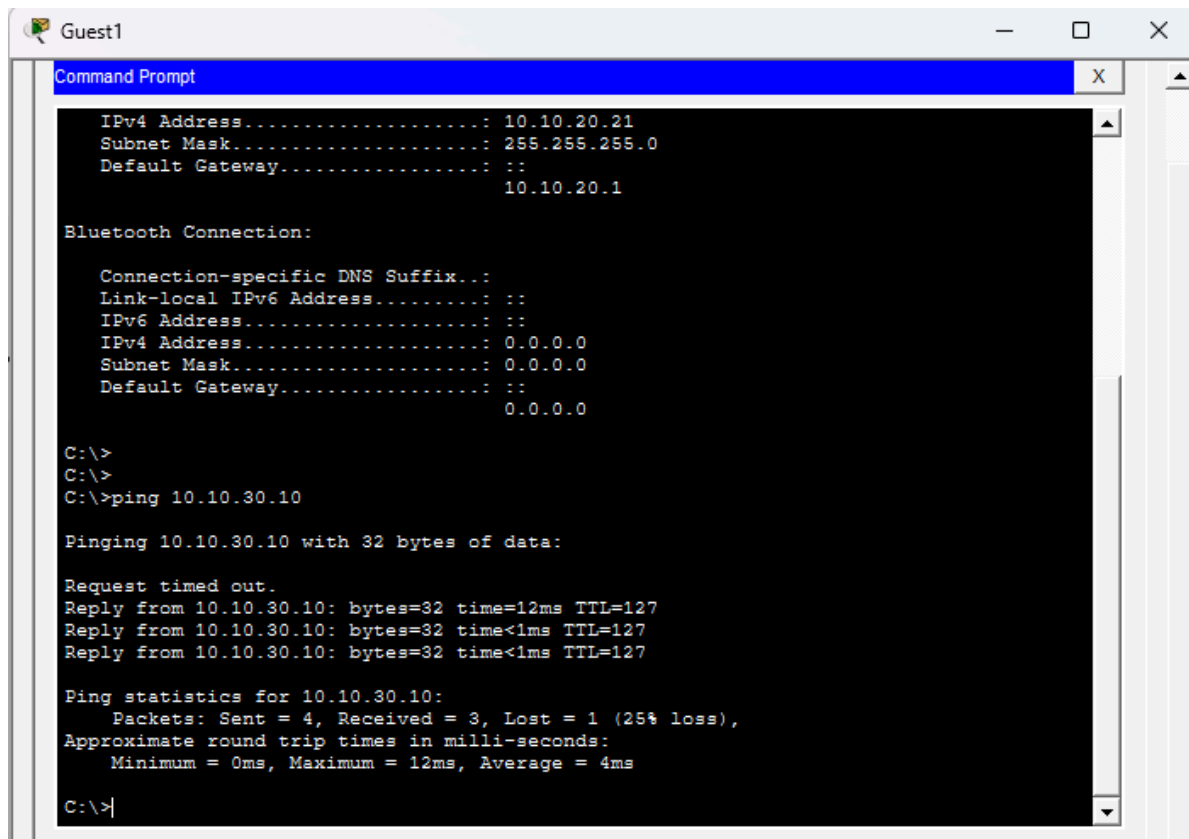
The final permit line is needed because every ACL ends with an implicit deny all.

5.3 Expected results

Test	Expected	Why
Guest 1 → Server	Fail	Matches the deny line inbound on G0/0.20
Guest 1 → Staff 0	Success	Destination is not the Server VLAN
Guest 1 → gateway 10.10.20.1	Success	Destination is not the Server VLAN
Staff 0 → Server	Success	Source is not the Guest VLAN
Server → Guest 1	Fail	The guest's reply (10.10.20.x → 10.10.30.10) is denied inbound on G0/0.20

5.4 Issue encountered: ACL not blocking traffic

Symptom: After applying the ACL, Guest 1 could still ping the server.



```
Guest1
Command Prompt
IPv4 Address.....: 10.10.20.21
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 10.10.20.1

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: 0.0.0.0

C:\>
C:\>
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.30.10: bytes=32 time=12ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>
```

Figure 24: Guest 1 still reaching the server after the ACL was applied

Diagnosis: The ACL had been created as GUEST_BLOCKER but applied to the interface as GUEST_BLOCK. IOS accepts a reference to an ACL that does not exist without any error, and treats it as permit all, so the config looked correct while nothing was filtered.

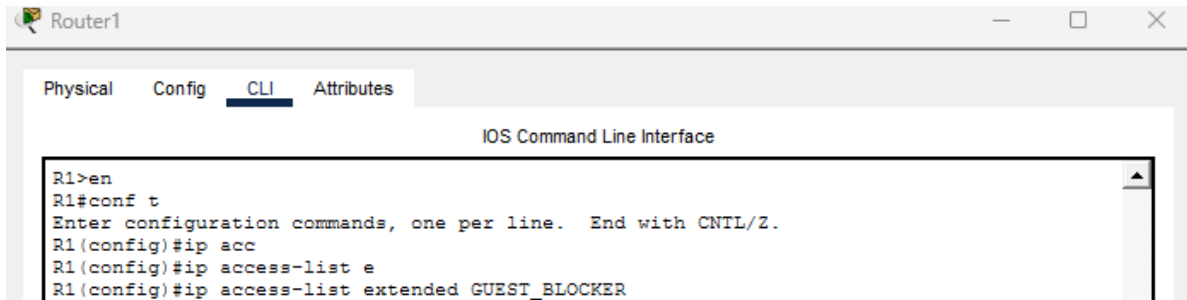


Figure 25: ACL created as GUEST_BLOCKER

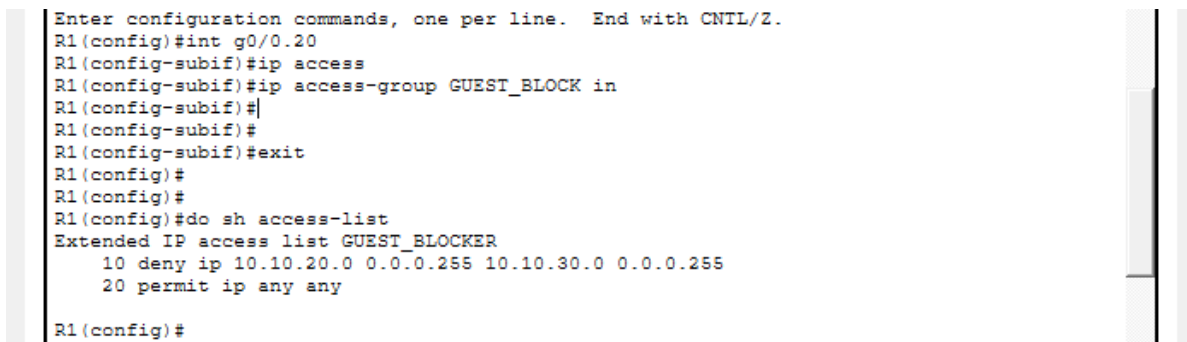


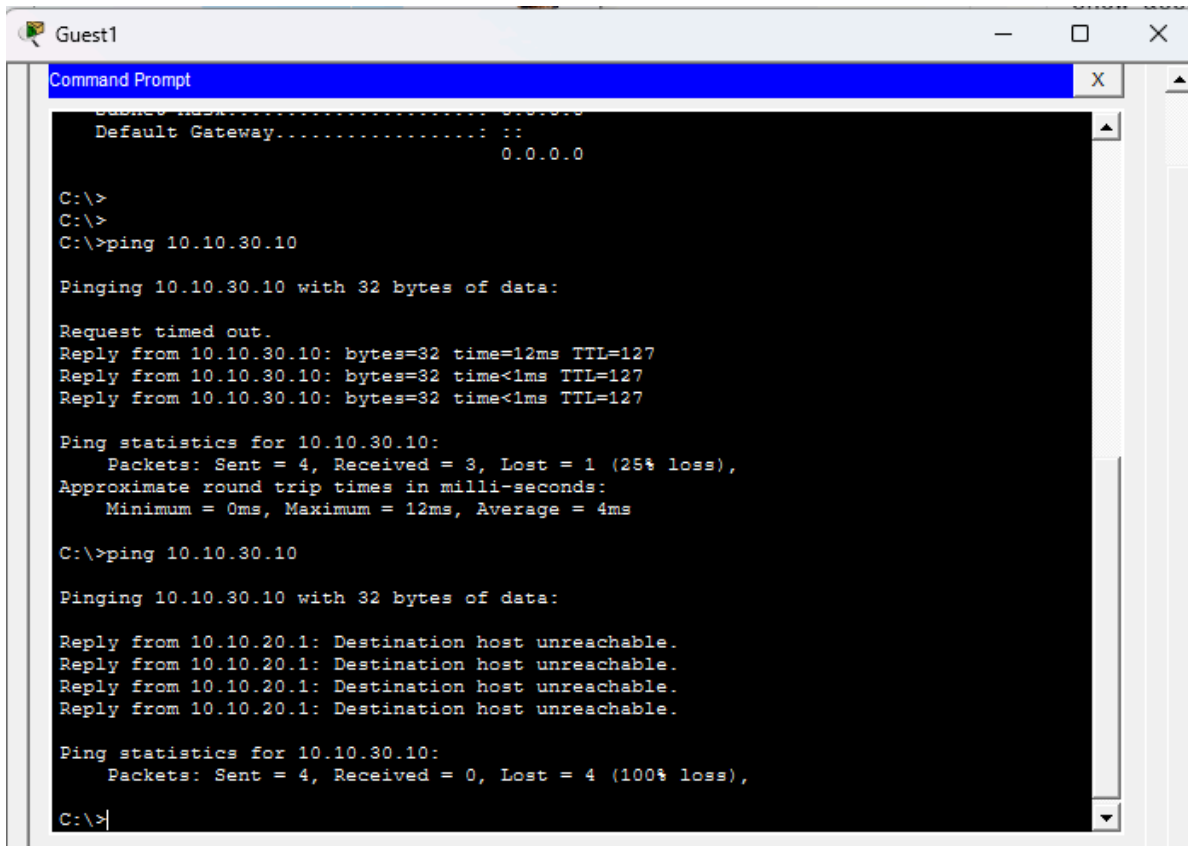
Figure 26: ACL applied to G0/0.20 as GUEST_BLOCK

Fix: Removed the wrong reference and applied the ACL with the correct name. Guest 1 → Server then failed as intended.

```
R1(config)# interface g0/0.20
R1(config-subif)# no ip access-group GUEST_BLOCK in
R1(config-subif)# ip access-group GUEST_BLOCKER in
```

Lesson: After applying an ACL, verify it with show ip interface and the match counters, not only with a ping.

5.5 Verification



```
Guest1
Command Prompt
C:\>
C:\>
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.30.10: bytes=32 time=12ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>ping 10.10.30.10

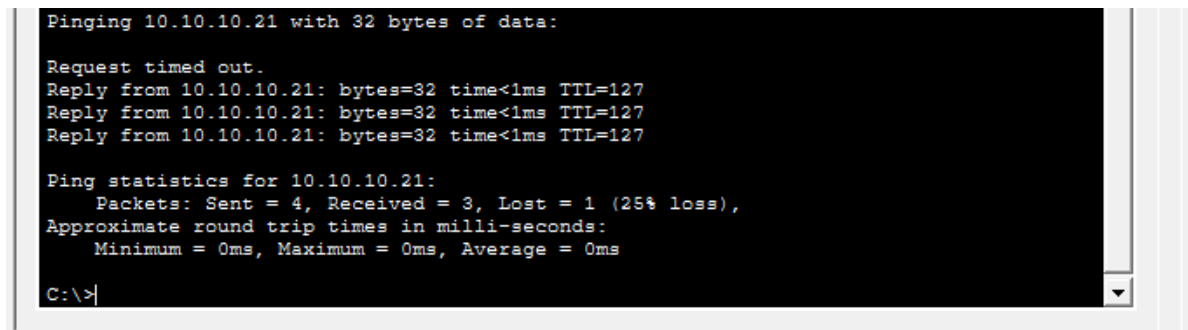
Pinging 10.10.30.10 with 32 bytes of data:

Reply from 10.10.20.1: Destination host unreachable.
Reply from 10.10.20.1: Destination host unreachable.
Reply from 10.10.20.1: Destination host unreachable.
Reply from 10.10.20.1: Destination host unreachable.

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Figure 27: Guest 1 → Server now blocked: R1 (10.10.20.1) replies Destination host unreachable



```
Pinging 10.10.10.21 with 32 bytes of data:

Request timed out.
Reply from 10.10.10.21: bytes=32 time<1ms TTL=127
Reply from 10.10.10.21: bytes=32 time<1ms TTL=127
Reply from 10.10.10.21: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.10.21:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

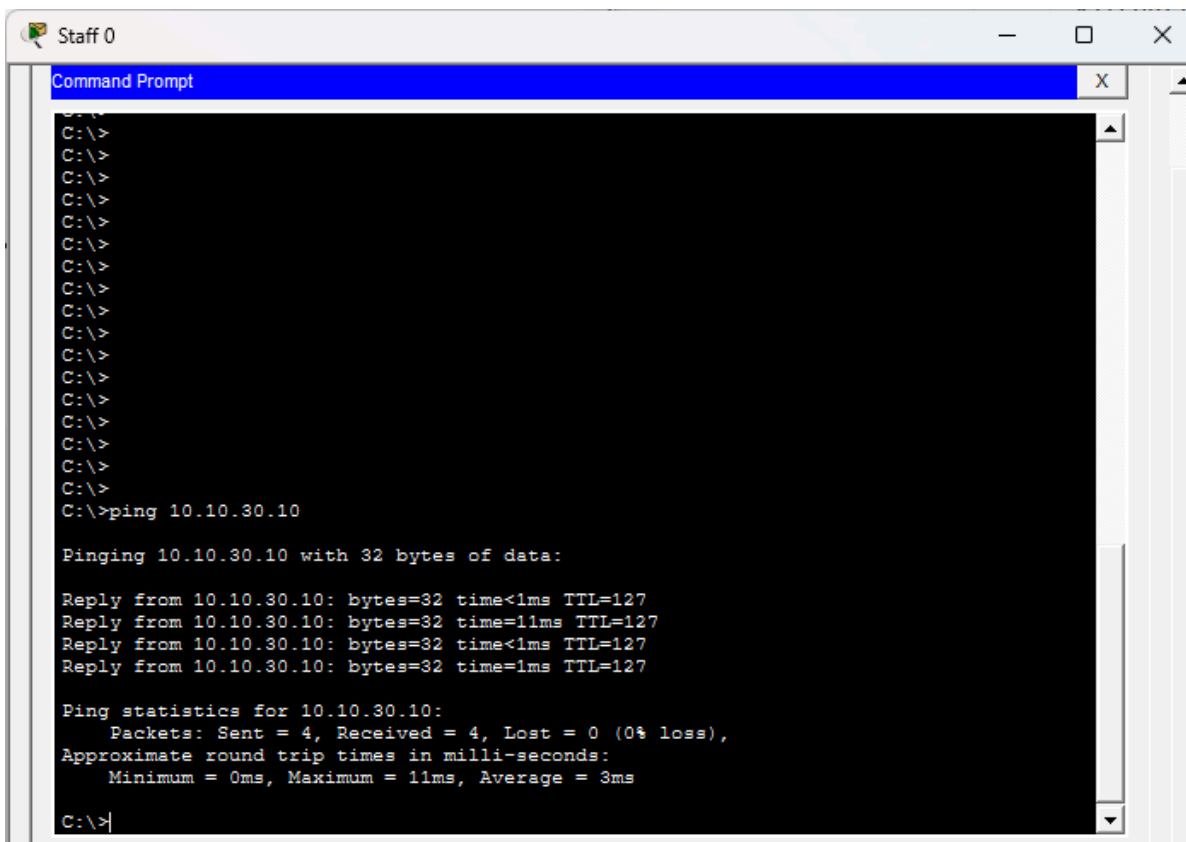
Figure 28: Guest 1 → Staff 0 still works

```
Pinging 10.10.20.1 with 32 bytes of data:
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 29: Guest 1 → gateway still works



```
Staff 0
Command Prompt
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=11ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

Figure 30: Staff 0 → Server still works

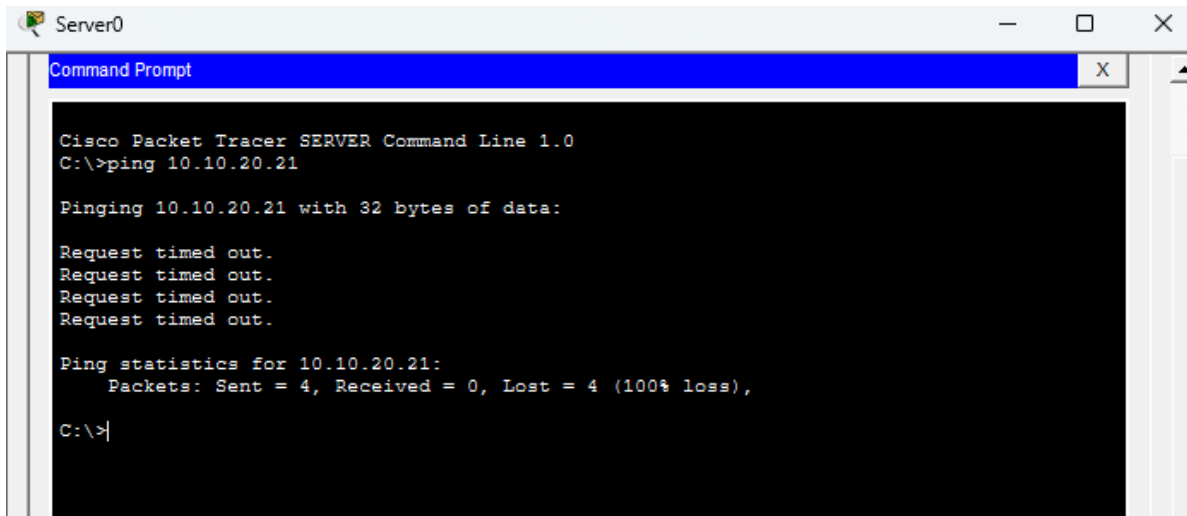


Figure 31: Server → Guest 1 fails, because the guest's reply is denied inbound on G0/0.20

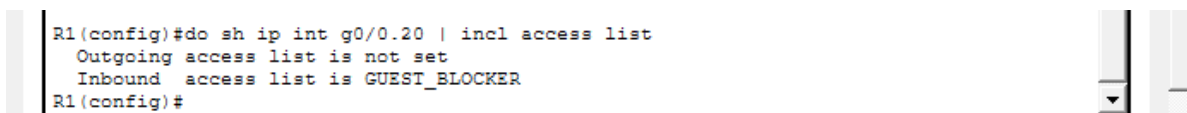


Figure 32: show ip interface g0/0.20 confirms GUEST_BLOCKER is applied inbound

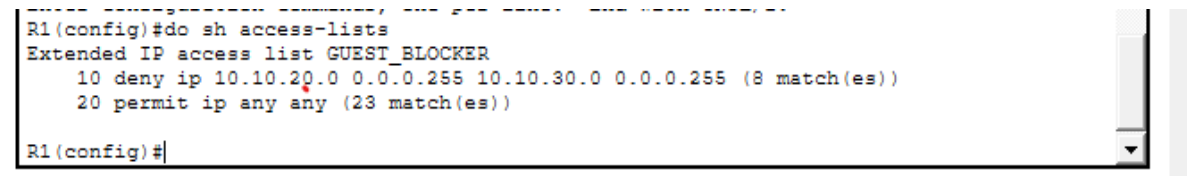


Figure 33: show access-lists incl. Deny and permit matches

Step 6: Secure Device Management (SSH)

6.1 Configuration

The same configuration was applied to R1, SW1 and SW2 (R1 shown):

```

R1(config)# ip domain-name harbourtech.local
R1(config)# username admin secret <password>
R1(config)# enable secret <password>
R1(config)# crypto key generate rsa
How many bits in the modulus: 2048

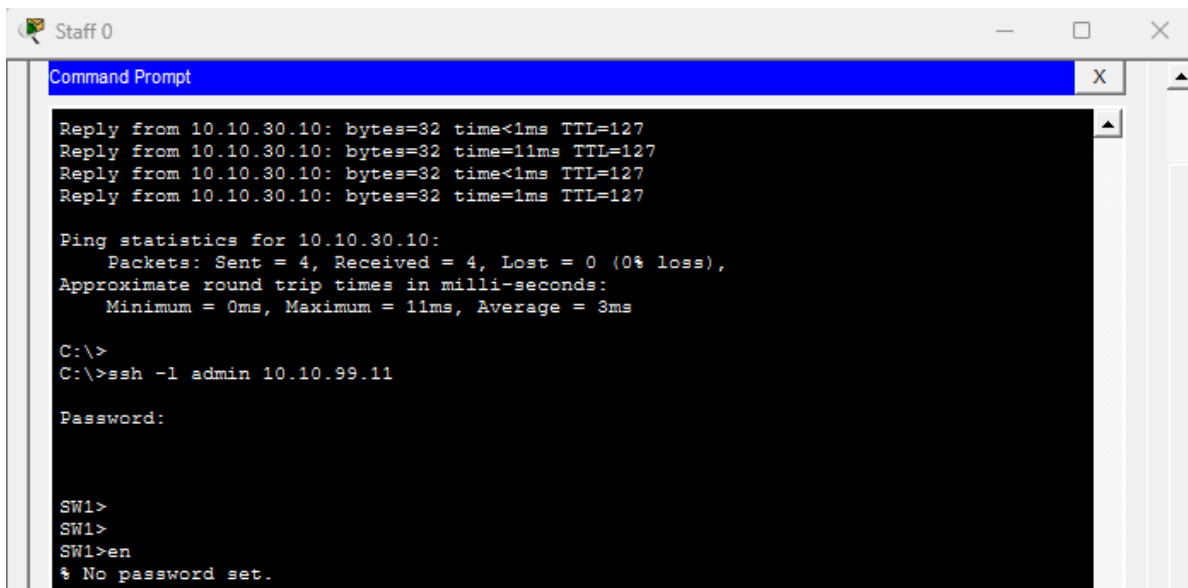
```

```
R1(config)# ip ssh version 2
R1(config)# line vty 0 15
R1(config-line)# login local
R1(config-line)# transport input ssh
```

- Domain name and hostname are required to generate the RSA keys
- 2048-bit keys are needed for SSH version 2
- login local uses the username stored on the device
- transport input ssh refuses Telnet, which sends everything in plain text
- enable secret stores the privileged-mode password as a hash

6.2 Issue encountered: enable refused over SSH

SSH login to SW1 worked, but enable returned "% No password set". IOS refuses privileged access over VTY lines when no enable secret is configured. Set an enable secret on all three devices.



```
Staff 0
Command Prompt
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=11ms TTL=127
Reply from 10.10.30.10: bytes=32 time<1ms TTL=127
Reply from 10.10.30.10: bytes=32 time=1ms TTL=127

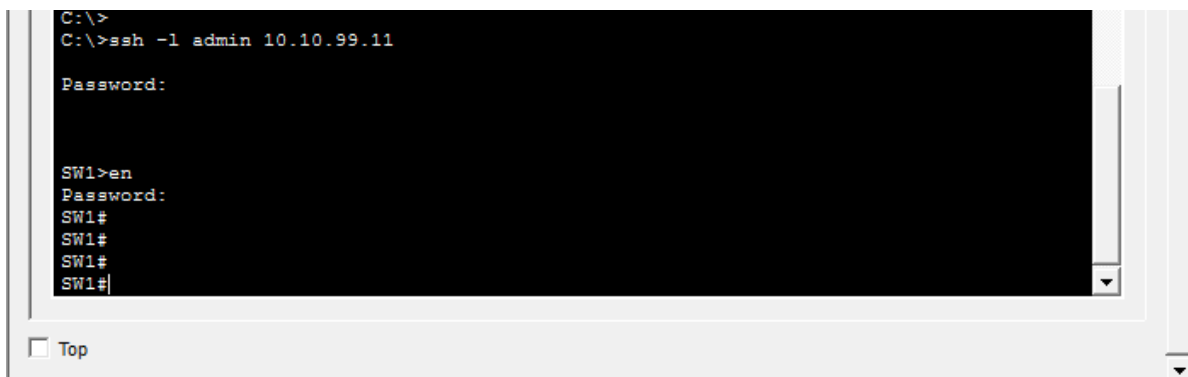
Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
C:\>ssh -l admin 10.10.99.11

Password:

SW1>
SW1>
SW1>en
% No password set.
```

Figure 34: SSH login works but enable is refused remotely



```
C:\>
C:\>ssh -l admin 10.10.99.11

Password:

SW1>en
Password:
SW1#
SW1#
SW1#
SW1#
```

Figure 35: After setting enable secret, privileged mode works over SSH

6.3 Verification (from Staff 0)

Test	Expected	Result
ssh -l admin 10.10.99.11 (SW1)	Login + enable	Success
ssh -l admin 10.10.99.12 (SW2)	Login + enable	Success
ssh -l admin 10.10.10.1 (R1)	Login + enable	Success
telnet 10.10.99.11 (SW1)	Refused	Connection closed by SW1

```
C:\>ssh -l admin 10.10.99.12

Password:

SW2>en
Password:
SW2#
```

☐ Top

Figure 36: Staff 0 SSH to SW2

```
C:\>ssh -l admin 10.10.10.1

Password:
% Login invalid

Password:

R1>
R1>en
Password:
Password:
R1#
R1#
```

Figure 37: Staff 0 SSH to R1 (first attempt was a mistyped password)

```
C:\>telnet 10.10.99.11
Trying 10.10.99.11 ...Open

[Connection to 10.10.99.11 closed by foreign host]
C:\>
```

Figure 38: Telnet to SW1: the TCP connection opens but SW1 closes it immediately because only SSH is allowed on the VTY lines

```
C:\>ssh -l admin 10.10.99.11

Password:

SW1>
SW1>
SW1>
SW1>
```

Figure 39: ssh from Guest 1 to SW1 - Guest can reach Management network - ACL does not cover it. Logged as security gap to fix

Step 7: Saving and Final Configuration

```
R1# copy running-config startup-config
SW1# copy running-config startup-config
SW2# copy running-config startup-config
```

The running configuration of each device was exported to R1.txt, SW1.txt and SW2.txt and is included with the project files.

Problems Encountered (Summary)

Problem	Cause	How it was found	Fix
Guest received a reserved address	GUEST DHCP exclusion missing from config	show run include excluded	Added exclusion, cleared bindings, renewed
ACL had no effect	ACL name typo on the interface (GUEST_BLOCK vs GUEST_BLOCKER)	show access-lists, show run interface	Re-applied with the correct name
enable refused over SSH	No enable secret configured	"% No password set" message	Set enable secret on all devices

Future Improvements

- Block Guest → MGMT (10.10.99.0/24) and restrict VTY access to the Staff subnet with an access-class, so guests cannot reach device logins
- Restrict trunks to the VLANs in use (switchport trunk allowed vlan 10,20,30,99,999)
- Migrate to VTPv3, or VTP transparent mode
- Port security on guest-facing access ports
- DNS service on the server, and internet access through NAT on R1
- Second branch site connected with OSPF
- Centralised DHCP on a server using ip helper-address

Project Files

- harbour-tech-v1.pkt
- configs/R1.txt, configs/SW1.txt, configs/SW2.txt
- This project log